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TABLEAU PROJECT 1

1. What is Tableau? Explain its importance in Business Intelligence and how it helps in data-driven decision-making.

- Tableau is a leading, user-friendly data visualization and business intelligence (BI) platform that transforms complex, raw data from multiple sources into interactive, easily understandable dashboards. It enables non-technical users to perform real-time analysis, identify trends, and drive data-driven decision-making across organizations.

Importance in Business Intelligence (BI)

- Data Connectivity: Connects to diverse sources like SQL, Excel, and cloud services (AWS, Azure).
- Ease of Use: Features a drag-and-drop interface that requires no advanced coding skills.
- Interactive Dashboards: Allows for real-time data exploration and visualization of complex, large datasets.
- Versatility: Used for reporting, forecasting, and data analysis across departments (e.g., finance, marketing).

Role in Data-Driven Decision-Making

- Actionable Insights: Converts raw data into visual insights, helping users spot patterns quickly.
- Real-time Decision Making: Facilitates faster,, accurate decisions based on current, rather than historical, data.
- Improved Collaboration: Enables sharing of interactive reports, facilitating informed, data-driven decisions company-wide.
- Predictive Analysis: Supports data modeling and forecasting to predict future trends.

2. Explain the role of the following Tableau components: a) Data Pane b) Worksheet c) Dashboard d) Story.

a) Data Pane (The Data Palette)

- **Role:** The Data Pane is located on the left side of the workspace and acts as the "palette" for your data. It lists all the data sources connected to the workbook and, within those, breaks down the data into fields.
- **Key Functions:**
    - Dimensions & Measures: It automatically categorizes data into Dimensions (qualitative fields like names, dates, or geographic data) and Measures (quantitative numeric fields that can be aggregated, such as sales or profit).
    - Field Management: Used to rename fields, change data types, create hierarchies, and create sets or calculated fields.
    - Drag-and-Drop: It is the primary source from which you drag fields onto shelves to build visualizations.

b) Worksheet (The Visualization Canvas)

- **Role:** A worksheet is the foundational canvas where you build individual data visualizations (charts, graphs, maps, or tables).
- **Key Functions:**
    - Analysis: This is where the actual data analysis takes place, allowing for iterative exploration of data.
    - Shelves and Cards: It provides shelves (Rows, Columns, Filters, Pages) and the Marks card to customize the visual representation (e.g., changing colors, sizes, and labels).
    - Foundation: Worksheets are the individual "building blocks" that are subsequently combined to create dashboards.

c) Dashboard (The Integrated View)

- **Role**: A dashboard is a central, interactive display that combines multiple worksheets (views) into a single, cohesive interface.
- **Key Functions:**
    - Comprehensive Insight: It allows users to monitor, compare, and analyze different, yet related, data points simultaneously.
    - Interactivity: Dashboards support actions like filters, highlights, and URL actions, allowing one chart to interactively update others.
    - Layout & Formatting: It provides layout containers and formatting tools to arrange, size, and style multiple visualizations for a polished, professional report.

d) Story (The Narrative Tool)

- **Role**: A story is a sequence of visualizations (worksheets or dashboards) that work together to convey a narrative, explain trends, or provide context to data.
- **Key Functions:**
    - Sequential Presentation: It operates like a slideshow, where each scene—called a story point—can represent a specific insight, filtering, or view.
    - Contextualization: It allows users to add annotations, text, and captions to guide the audience through a logical, data-driven story.
    - Narrative Drive: It helps move beyond mere data exploration to "telling a story" that explains the *why* behind the numbers.

3. What is the difference between Dimensions and Measures in Tableau? Provide examples of each.

- In Tableau, Dimensions are qualitative, categorical data (e.g., names, dates, regions) used to categorize and segment data, defining the level of detail. Measures are quantitative, numerical data (e.g., sales, profit, quantity) that are aggregated (sum, average, count). Dimensions create headers, while Measures create axes.

Examples in Tableau

Dimensions (Blue Fields - Top of Data Pane):

- Customer Name: Used to group sales by individual customers.
- Order Date: Used to segment data by Year, Quarter, or Month.
- Region/City: Used to categorize data geographically.
- Product Category: Used to break down sales into, e.g., Furniture, Office Supplies.

Measures (Green Fields - Bottom of Data Pane):

- Sales: Aggregated as `SUM(Sales)` to show total revenue.
- Profit: Calculated as `AVG(Profit)` to see profitability.
- Quantity: Counted or summed to show volume.
- Discount: Average discount percentage.

4. Define and explain the purpose of Filters, Parameters, and Sets in Tableau.

Filters, Parameters, and Sets in Tableau are tools used to control, analyze, and manipulate data visualization. Filters restrict data to specific dimensions or

measures, Sets create custom subsets of data based on conditions, and Parameters are workbook-wide variables that replace constant values to add interactivity, such as dynamic Top N filtering or metric switching.

Filters

- Definition: A mechanism to include or exclude specific data points (dimensions or measures) from a visualization.
- Purpose: To narrow down data, focus on specific regions/categories, or create interactive dashboards where users can filter by date, category, etc..
- Types: Dimension Filters, Measure Filters, Data Source Filters, and Context Filters.

Parameters

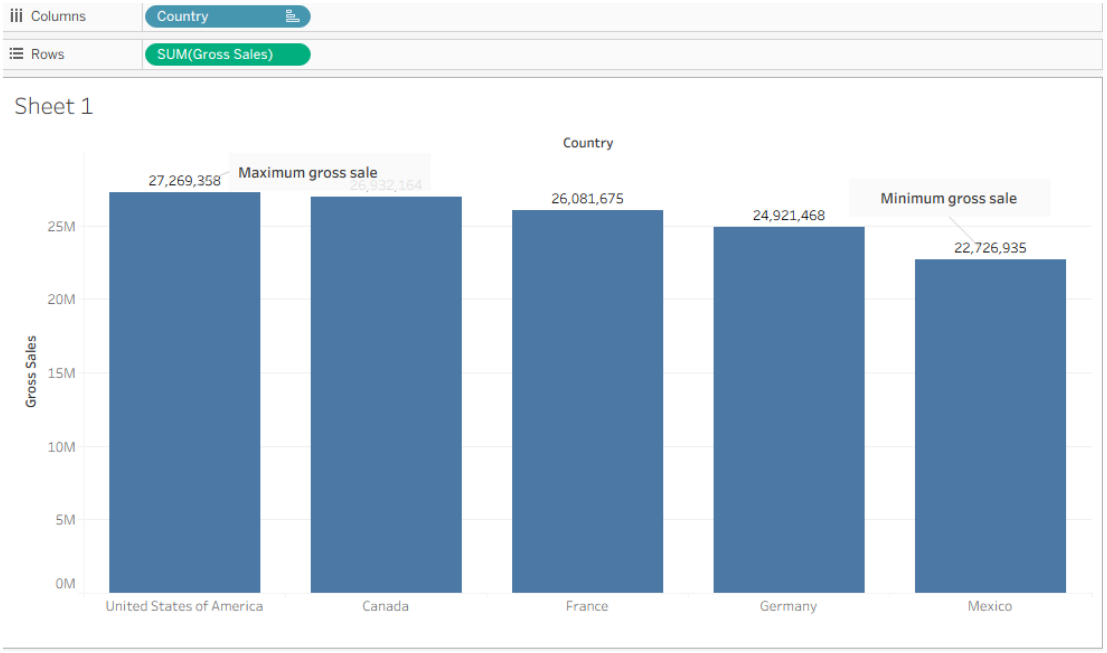
- Definition: User-input variables (numbers, dates, or strings) that replace constant values in calculations, filters, or reference lines.
- Purpose: To add interactivity, allowing users to dynamically change inputs without editing the underlying data structure (e.g., changing a Top 10 list to Top 20).
- Key Features: Workbook-level scope (affects multiple sheets), single-select, and used with calculated fields.

Sets

- Definition: Custom fields that define a subset of data based on specific conditions or manual selection.
- Purpose: To compare subsets against the whole, such as comparing "Top Customers" versus "All Other Customers" in a view.

- Types: Fixed Sets (static selection) or Dynamic Sets (update based on data changes)

5. Create a bar chart showing Gross Sales by Country. ● 🔗 Dataset Link:Global_sales_dataset ● Sort the countries in descending order of sales ● Highlight or annotate the bar that represents the maximum and minimum Gross Sales. ● Add data labels and format the chart for presentation.



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6. Using Tableau, create a dual-axis chart that displays: ● 🔗 Dataset Link:Global_sales_dataset ● Monthly Sales as bars ● Monthly Profit as a line ● Filter the data to include only records from the year 2014 ● Ensure both axes are synchronized and properly labeled ● Add an appropriate chart title, and format the chart for clear visual presentation ● Paste a screenshot of the final chart in your submission

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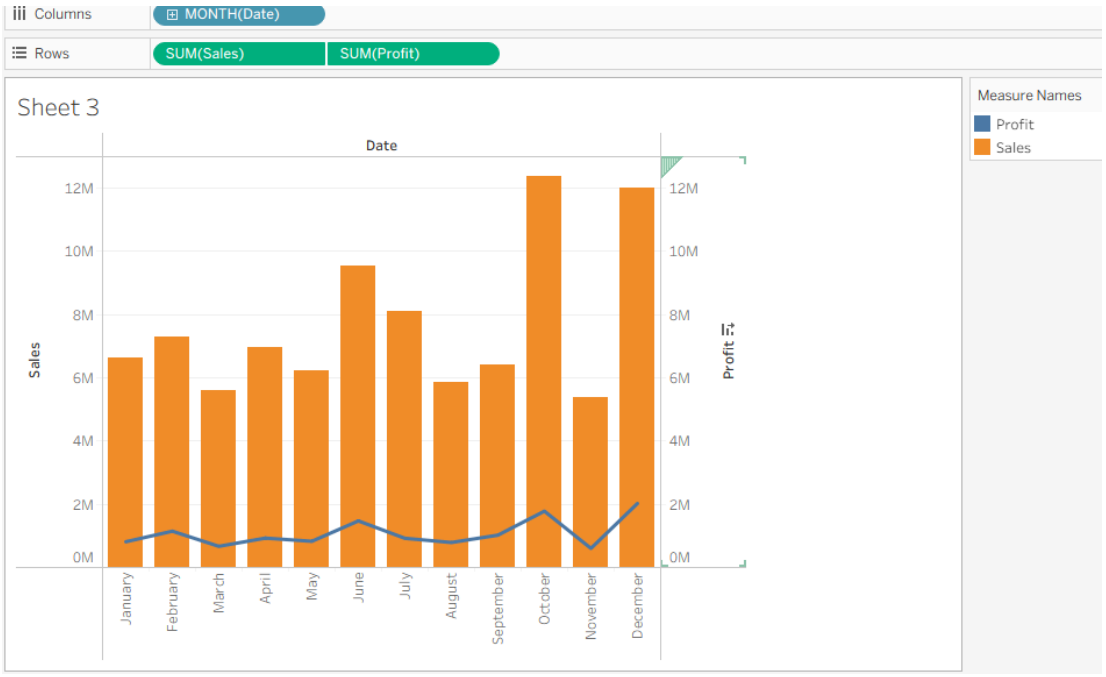

7. Create a filled map showing total Units Sold by Country. ● 🔗 Dataset Link:Global_sales_dataset ● Add a parameter to allow users to switch between Units Sold and Profit. ● Use the Discount Band as a filter in your visualization.

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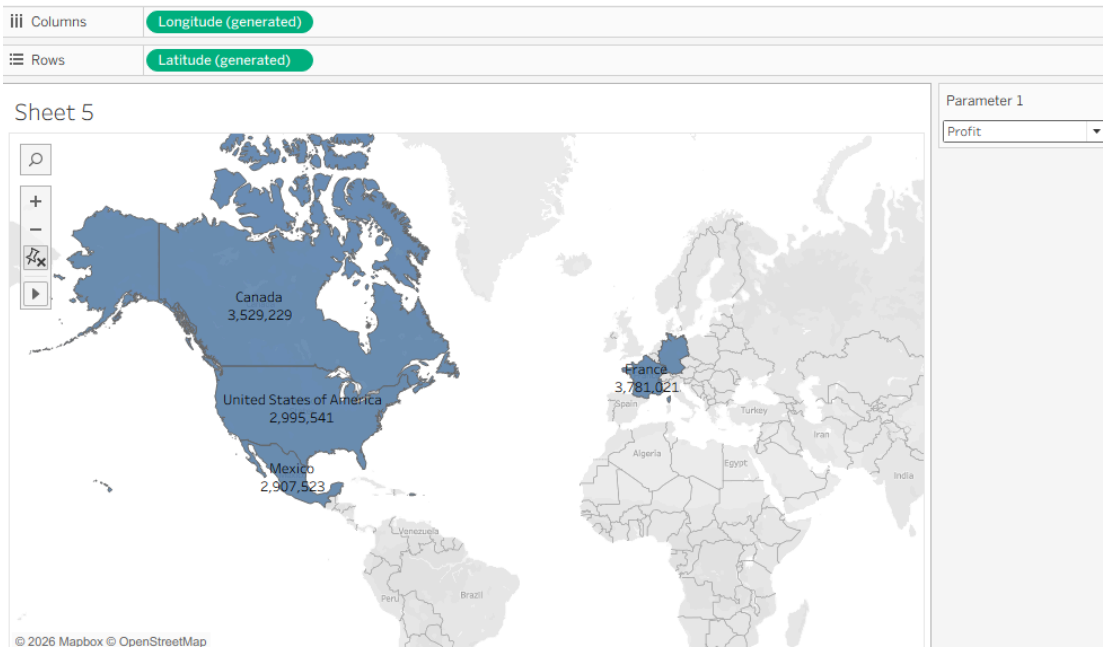

8. Create a dashboard that includes: ● 🔗 Dataset Link:Global_sales_dataset ● KPI tiles for Total Sales, Total Profit, and Total Units Sold ● A line chart for Profit trend over time ● Filters for Product and Country Ensure your dashboard is interactive and visually appealing

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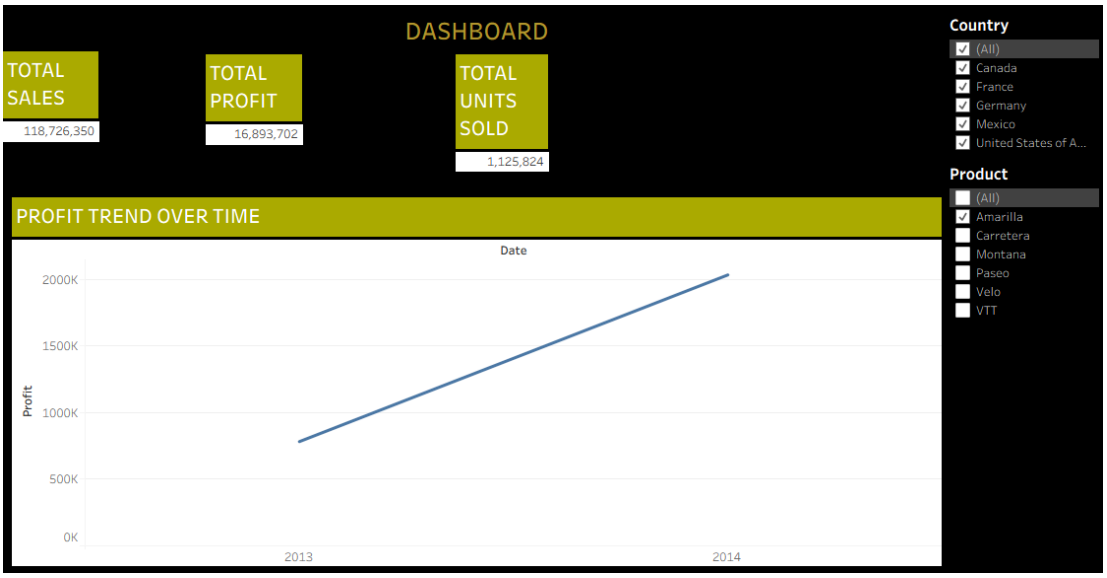

9. Your goal is to identify products that generate low profit despite high sales volume. ● 🔗 Dataset Link:Global_sales_dataset ● Use scatter plot or highlight table to identify such products. ● Add filters for Country and Segment. ● Write two business insights based on your chart.

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INSIGHTS:

- The scatter plot shows that the most high sale products aslo generate high profit. However, Amarilla generates relatively high sales but does not perform as strongly in profit compared to Paseo, indicating possible margine optimization opportunities.
- When filtered by Country and Segment, profitability distribution changes, showing that canada Vtt is high on sale but low in profit and other counties vary slightly.

10. [Scenario-Based – Customer Behavior & Retention Strategy] Dataset to Use: online_retail_II Dataset Name: Online Retail II Dataset Source: UCI Machine Learning Repository – Online Retail II Dataset Business Scenario: You are a Data Analyst at an e-commerce company that sells home decor and gifts across multiple countries. The leadership team is concerned about customer churn and revenue loss due to inconsistent customer behavior. They've asked

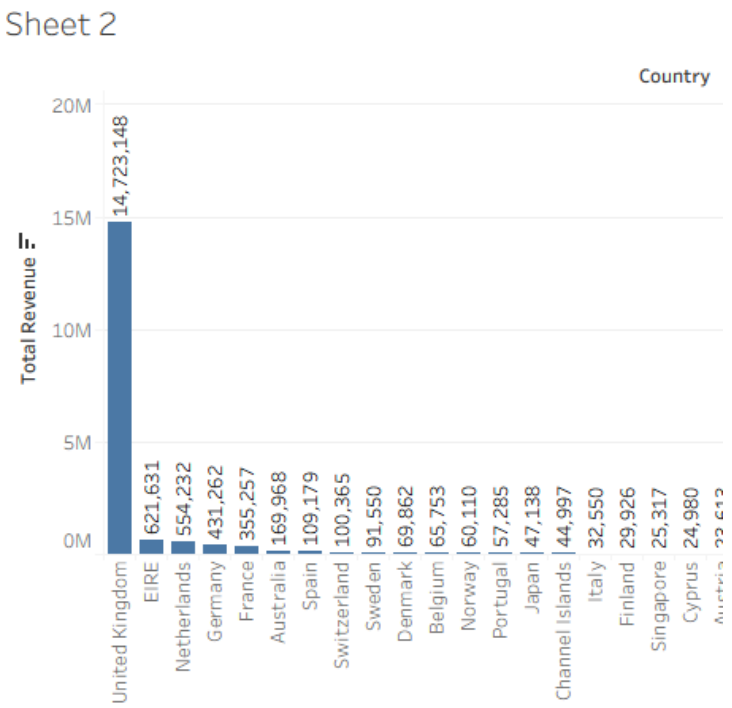
you to investigate patterns in customer orders, returns, and geographic sales performance from the Online Retail II dataset. Your Task in Tableau:

1. Use Tableau to answer these questions:

○ **Which countries have the highest number of repeat customers.?**

- United kingdome has the highest number of repeat customer.

○ **What is the return rate by product and Find top 10 countries?**



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○ **What time of year do customers tend to buy the most (Seasonality)** ?

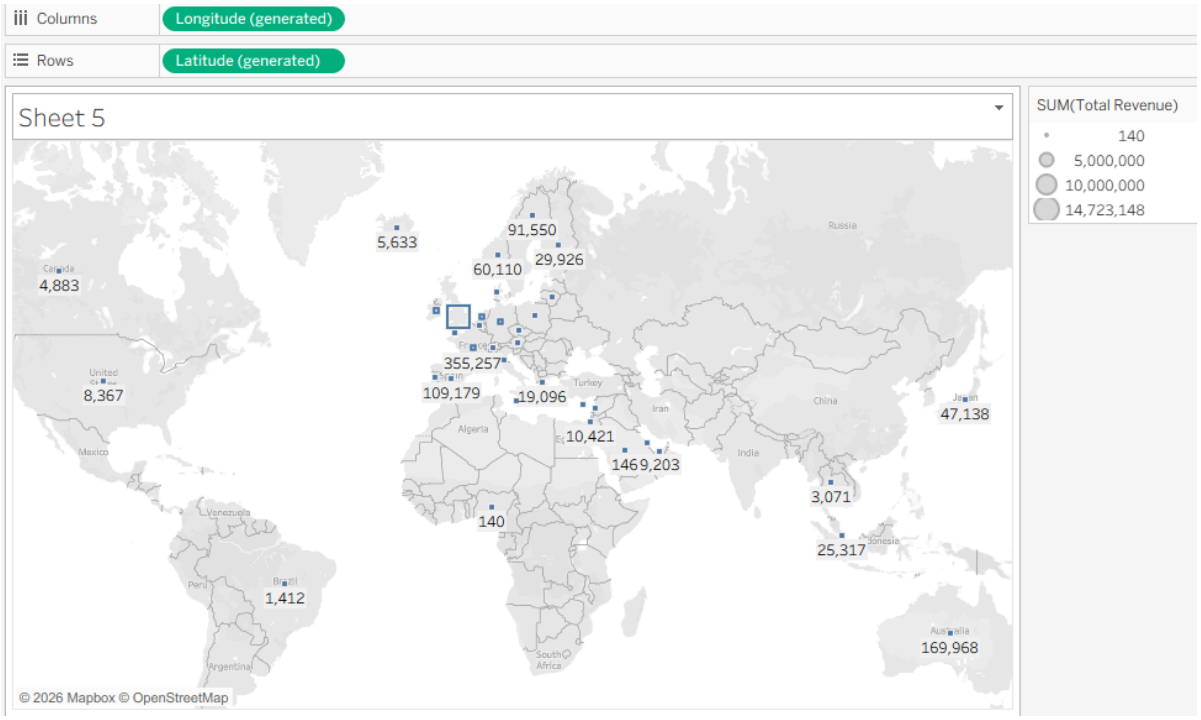
- Most likely peak: November–December (Christmas season 🎄)

○ **Are there certain customers with high order value but also high return rates?**

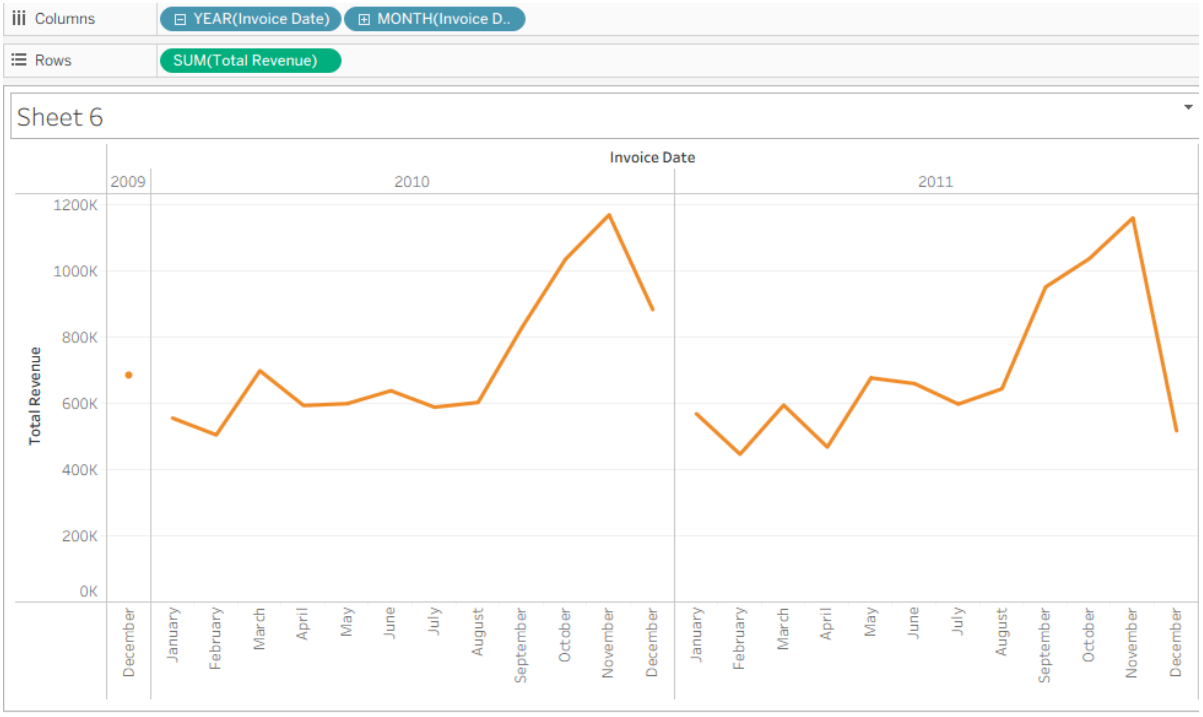
- These customers contribute revenue but also high return losses.

2. Create visualizations:

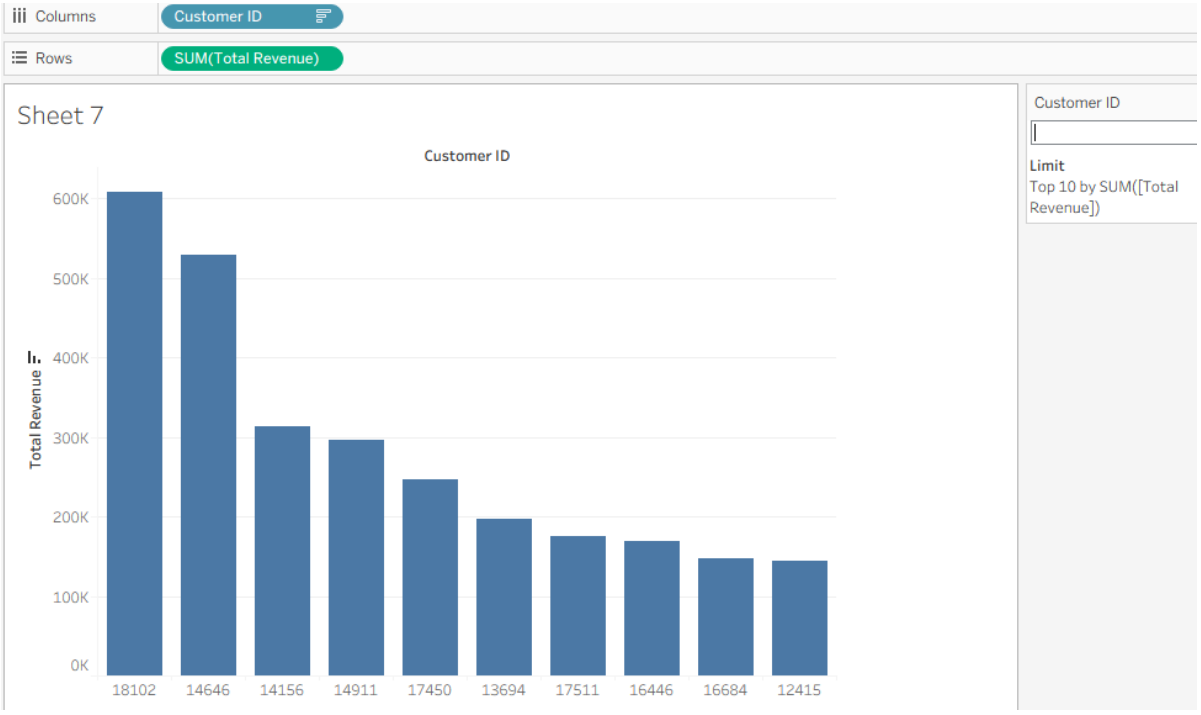
○ **A map showing Revenue by Country**



○ **A line chart of Monthly Sales Trend**



○ **A bar chart showing Top 10 customers by Total Revenue**



○ **A table/heatmap showing Top returned products by country**

Columns: Description

Rows: Country

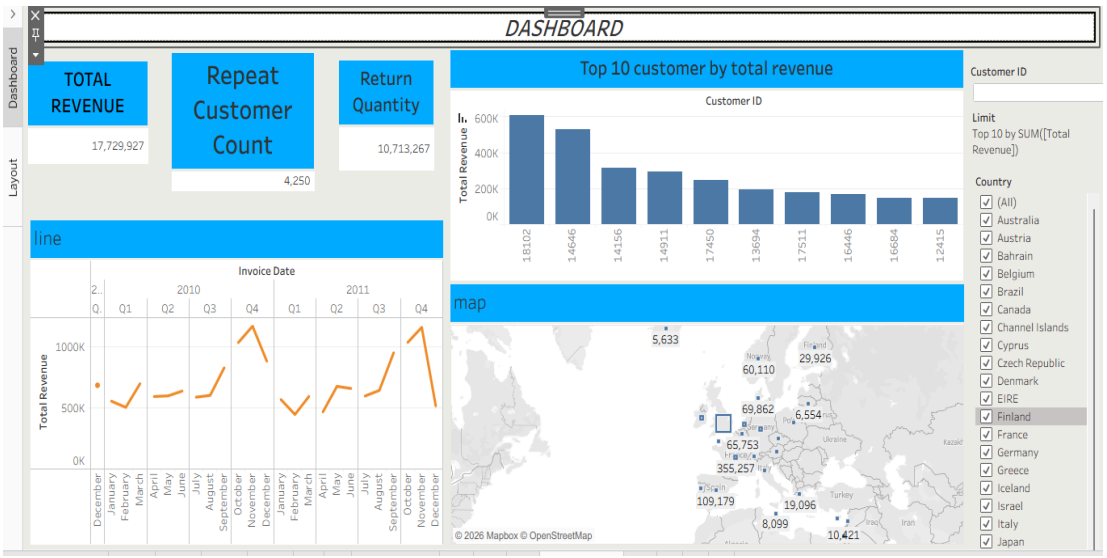
Sheet 8

	Description									
Country	ASSORTED ..	HOME BUIL..	JUMBO BA..	LUNCH BAG ..	LUNCH BAG..	PARTY BUN..	REGENCY C..	REX CASH+..	STRAWBER..	WHITE HAN..
Australia			419	294	1,002	1,174	2,931			18
Austria	230		59		17	40	51		60	
Bahrain							26		15	
Belgium	14		321	17	314	63	676		30	
Brazil	10				17		175			9
Canada	14			83	50		13		30	18
Channel Islands	68	283	956	99	99	347	786			372
Cyprus	284	42	21		17	33	950	1	38	782
Denmark				17	33	59			213	
EIRE	608	631	376	170	363	2,820	15,067		844	4,729
Finland		36	183			59	102			71
France	3,296	232	1,454	417	1,393	1,039	3,518	6	438	250
Germany	541	126	1,638	284	710	389	13,600	1	1,386	159
Greece	95			17	17		213		15	408
Iceland							230			
Israel		18				5	726	1		6
Italy	41	89	21		50		128		64	204
Japan						676				
Lebanon		36				70	153			
Malta							153		15	18
Netherlands	1,651		5,613	1,541	2,599	667	6,871		30	10,933
Nigeria								1		
Norway	27	18	179		50	59	370		60	
Poland	14								316	
Portugal	270	54	400		234		618		120	71
Singapore			61			40	927		15	380
Spain	1,628	298	84	500	25	422	2,521	5	111	378
Sweden		18	40	870			13		30	18
Switzerland	81	36	411	50	132	761	937		60	386
United Arab Emirates				17	17	47	242		75	
United Kingdom	118,167	41,905	124,702	36,214	23,330	95,094	234,186	3,899	36,636	232,829
USA			21				204		15	

3. Build a dashboard for business insights:

○ **Allow filters for Country, Product, and Customer ID**

○ **Use KPIs for: ■ Total Revenue ■ Total Returns ■ Repeat Customer Count**


DASHBOARD
Dashboard
Layout
TOTAL REVENUE
17,729,927
Repeat Customer Count
4,250
Return Quantity
10,713,267
Top 10 customer by total revenue
Customer ID
Total Revenue
600K
400K
200K
0K
18102
14646
14156
14911
17450
13694
17511
16446
16684
12415
line
Invoice Date
2010
2011
Q1
Q2
Q3
Q4
Total Revenue
1000K
500K
0K
December
January
February
March
April
May
June
July
August
September
October
November
December
map
5,633
29,926
60,110
69,862
6,554
65,753
355,257
109,179
19,096
8,099
10,421
© 2026 Mapbox © OpenStreetMap
Customer ID
Limit
Top 10 by SUM([Total Revenue])
Country
(All)
Australia
Austria
Bahrain
Belgium
Brazil
Canada
Channel Islands
Cyprus
Czech Republic
Denmark
EIRE
Finland
France
Germany
Greece
Iceland
Israel
Italy
Japan